

Tiffany Yu

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EDUCATION

University of South Carolina

Bachelor of Science in Computer Engineering

Columbia, SC

Expected Graduation: December 2026

GPA: 3.569

SKILLS

Technical: Programming Languages | Robotic systems and tools | Communication Protocol – UDP, TCP | Network edges | AutoCAD | Graphic Design | Spice | LabVIEW

Operating Systems: Window 10&11 | Ubuntu | Linux | Android

Programming: Python, Java, Git Repository, C++, Docker, ROS, MATLAB, LaTeX, Assembly Language

Languages: Fluent in Chinese, Beginner in Spanish

WORK EXPERIENCE

University of South Carolina

Columbia, SC

Discrete Structures Math Grader

Dates: August 2025 – Current

- Graded assignments, homework, and quizzes, in Discrete Structures, ensuring consistency, accuracy, and timely feedback for students.
- Provided written feedback that helped students improve reasoning, proof structure, and conceptual understanding.
- Managed grading workload under deadline pressure, ensuring high quality and fairness while maintaining confidentiality.

Undergraduate Research Assistant

Dates: December 2024 – Current

- Conducted research on network edge structures and cost optimization, analyzing performance trade-offs in hardware and software implementations.
- Developed and implemented a Random Forest algorithm to prune random graphs of nodes and edges, improving computational efficiency and scalability.
- Collaborated on coding a software solution to simulate and optimize network architectures, leveraging advanced deep machine learning techniques.

Undergraduate Research Assistant

Dates: March 2024 – January 2025

- Assisted with setting up and maintaining underwater robotic equipment, ensuring proper function during research experiments.
- Help organize and prepare lab space and equipment before experiments, improving workflow efficiency for the team.

RESEARCH INVOLVEMENT

International Conference on Computing and Machine Intelligence

Dates: April 9, 2026

Machine Learning-Based Graph Simplification for Symbolic Accelerators

Al Ahsa, Saudi Arabia (Online)

SuperComputing International Conference

Dates: November 18, 2025

AutoSlim: Intelligent Automata Graph Optimization for Efficient Acceleration

St. Louis, MO

University of South Carolina

UOFSC Summer Symposium

Dates: July 31, 2025

ML-Based Automata Simplification for Symbolic Accelerators

Columbia, SC

UOFSC Discover Day

Dates: April 25, 2025

CSE Graduate Research Symposium

Dates: March 7, 2025

Enhancing Security and Efficiency in Computational Hardware Through Intelligent Optimization

RESEARCH PAPERS

Yu, T., Stahle-Smith, R., Eswaramoorthi, D., Karakchi, R. (2025). *ML-Based Automata Simplification for Symbolic Accelerators*. arXiv preprint arXiv:2507.08751.

Karakchi, R., Stahle-Smith, R., Chinnasami, N., **Yu, T.** (2025). *Toward a Lightweight, Scalable, and Parallel Secure Encryption Engine*. arXiv preprint arXiv:2506.15070.

LEADERSHIP & INVOLVEMENT

University of South Carolina

Columbia, SC

University 101 Peer Leader

Dates: Fall 2025

- Managed difficult conversations and provided peer support, enhancing communication skills and conflict resolution.
- Led classroom co-instruction, facilitating discussions and speaking in front of peers to guide academic success.

Vice President of Finance at Alpha Lambda Delta

Dates: Spring 2025 - Current

- Managed financial records and budgets for the honor society, ensuring accuracy and accountability in all transactions.
- Oversaw collection and allocation of membership dues, supporting organization operations and events.
- Coordinated with campus offices and vendors to process payments and reimbursements in compliance with university policies.

Peer Mentorship Program

Dates: Fall 2024

- Mentored first-year students by offering academic guidance and support, helping them navigate university life.
- Organized regular check-ins with mentees to track their progress, providing resources and advice tailored to their individual needs.
- Acted as a role model for mentees, sharing personal experiences to inspire confidence and success in academic pursuits.

Capstone Scholar Leadership Program

Dates: Spring 2024

- Participated in in-depth discussions and coursework on leadership theories and practices, enhancing understanding of leadership dynamics.
- Developed and presented a group project, applying leadership strategies to successfully complete tasks within deadlines.

Awards

University of South Carolina

Columbia, SC

Dean's Honor List

Dates: Spring 2025, Fall 2024

Capstone Fellowship Distinction

Dates: Fall 2025

President's Honor List

Dates: Fall 2023

Capstone Scholar

Dates: Fall 2023 – Spring 2025